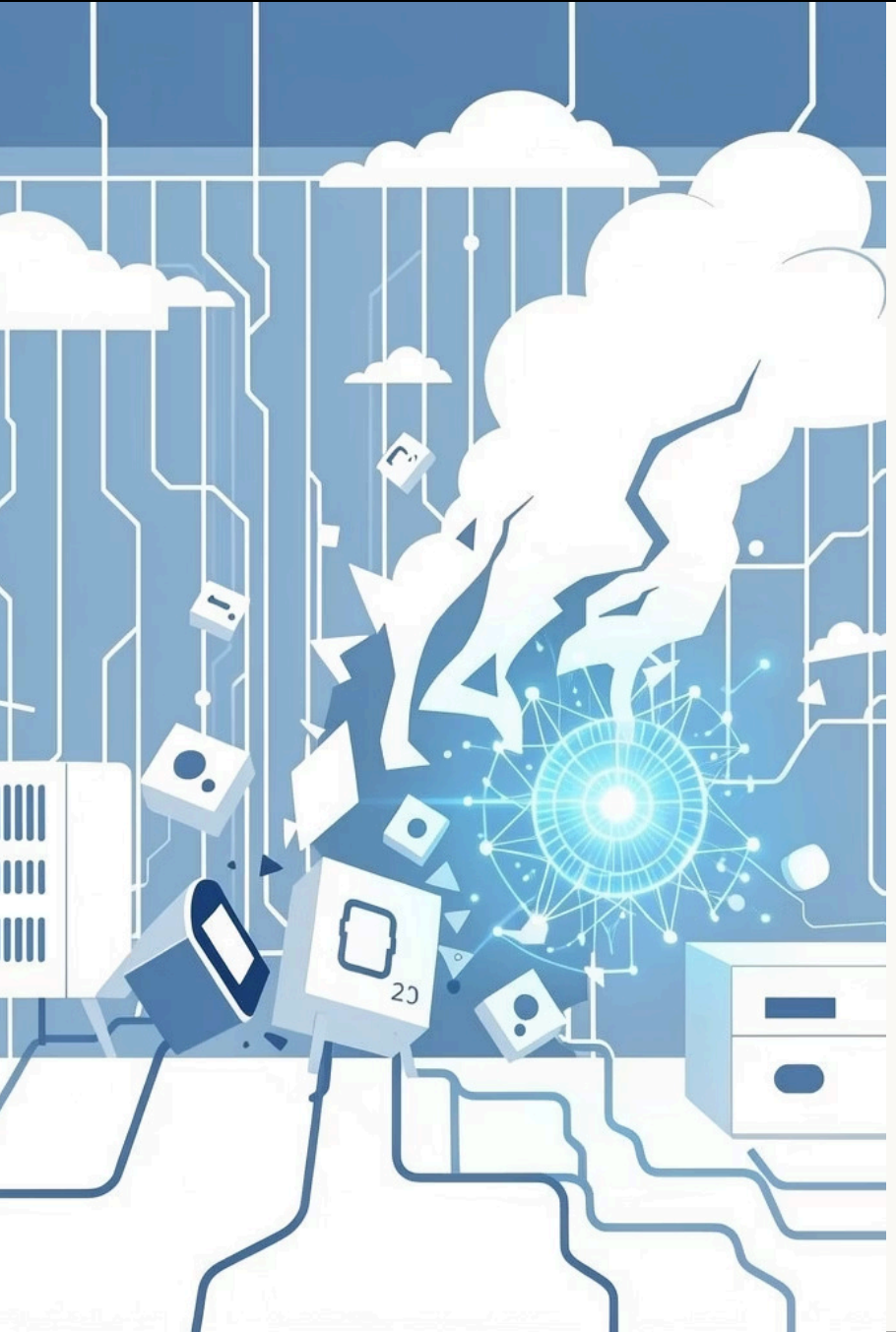


AEGIS AI

CyberShield 2026

Autonomous cyber-immune platform

Team: Algorithm Avengers

Sai Vijay Ragav · Jaswanth Prasanna · Ranjith Kumar · Nitheesh Kumar

The Cybersecurity Problem

Evolution Outpaces Defense

Threats evolve faster than signature-based defenses, leaving gaps for new attack techniques.

Zero-Day Vulnerabilities

Unknown exploits bypass static rules and signature lists.

Alert Fatigue

Noisy alerts and manual triage overwhelm teams, increasing missed threats.

Slow Mitigation

Manual response workflows delay containment, widening impact.

Vision: Digital Immune System

Like a human immune system, **AEGIS** learns normal digital behavior, spots anomalies, and responds automatically to protect systems.

Continuous Monitoring

24/7 baseline of users, devices, and traffic to detect deviations.

Adaptive Detection

Models learn and update to reduce false positives and catch subtle threats.

Automated Response

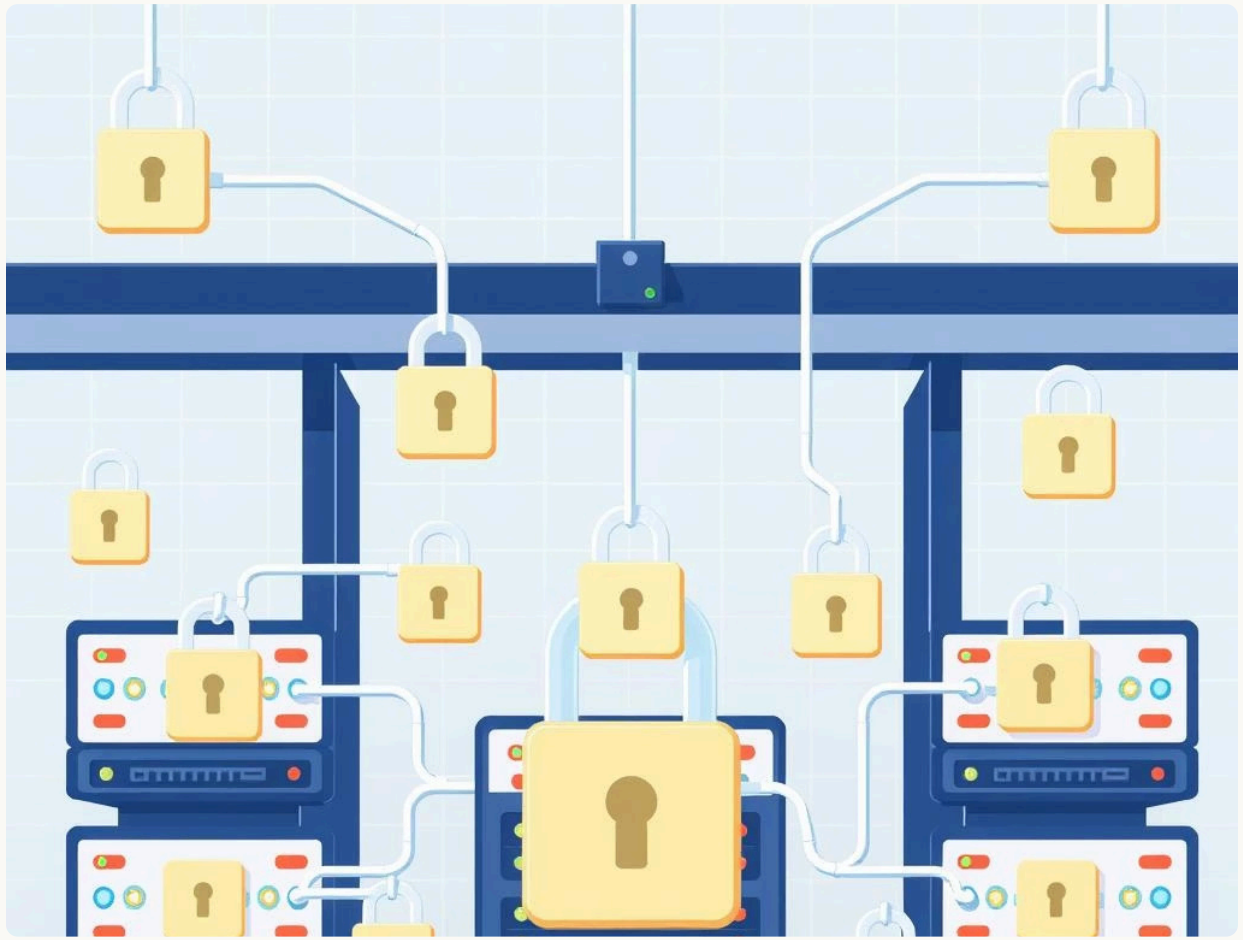
Immediate containment and remediation to restore normalcy without manual delay.

Turn Security into Business Resilience

AEGIS AI

AEGIS AI is an autonomous cyber defense platform that transforms security from a cost center into a strategic enabler. It combines AI detection with automated response to reduce risk and keep operations running.

- Reduce breach impact — contain threats automatically to cut downtime and losses.
- Lower ops cost — fewer false alerts and less manual triage.
- Protect continuity — rapid automated remediation for critical systems.
- Enable confident decisions — explainable AI for execs and auditors.



Key Innovations

Threat DNA Fingerprinting

Converts attacks into behavioral "DNA" vectors across network, endpoint, and user telemetry. Detects mutated threats by matching behavior instead of static indicators.

Adaptive Honeytokens

Context-aware decoy credentials, files, and endpoints that evolve with the environment. Generate high-fidelity alerts and forensic trails with minimal false positives.

Autonomous Defense Playbooks

AI-ranked playbooks that automatically observe, isolate, remediate, and validate actions — with human override and audit logs for compliance.

Explainable AI

Clear threat rationales showing features, evidence chains, and confidence scores to speed analyst validation and triage.

Future Enhancements: Next-Generation Capabilities

AEGIS is evolving — roadmap items add advanced AI and collective intelligence to outpace sophisticated threats.

Dark Web Intelligence

Proactively monitor dark web sources to spot emerging threats and leaked credentials.

Federated Threat Learning

Share anonymized threat signals across partners to improve global detection.

AI-Driven Simulations

Run adaptive AI attack simulations to test and strengthen defenses.

Self-Healing Infrastructure

Automate remediation so systems recover integrity with minimal human input.

AEGIS: Our Core Competitive Advantages

AEGIS unifies advanced technologies into an intelligent defense platform for superior threat detection and response.

Threat DNA Fingerprinting

Detects evolving attack variants by recognizing unique behavioral signatures.

Adaptive Honeytokens

Delivers high-fidelity alerts via dynamic, context-aware deception.

Explainable AI

Provides clear reasoning for detections to speed analyst validation.

Autonomous Response

Automatically contains and remediates threats using AI playbooks.

AEGIS Technology Stack

Scalable stack that powers AEGIS's real-time threat detection and automated response.

Backend

Python + FastAPI — high-performance, async APIs for core security logic.

Machine Learning

Scikit-Learn, **XGBoost**, **Isolation Forest**, **SHAP** — modeling, anomaly detection, explainability.

Streaming & Data

Redis Streams — real-time ingestion and processing for immediate analysis.

Frontend

React, **Streamlit** — interactive, intuitive dashboards.

Infrastructure

Docker — containerized deployments for scalability and consistency.

Mobile

Java, **Kotlin** — mobile application development.

Tangible Impact: Measuring AEGIS's Value

AEGIS cuts detection and response times while lowering false positives.

<1

MTTD

From hours to seconds.

<1

MTTR

From hours to milliseconds.

95%

False Positives

~95% reduction via behavioral models.

Result: lower breach costs, better efficiency, and stronger security posture.